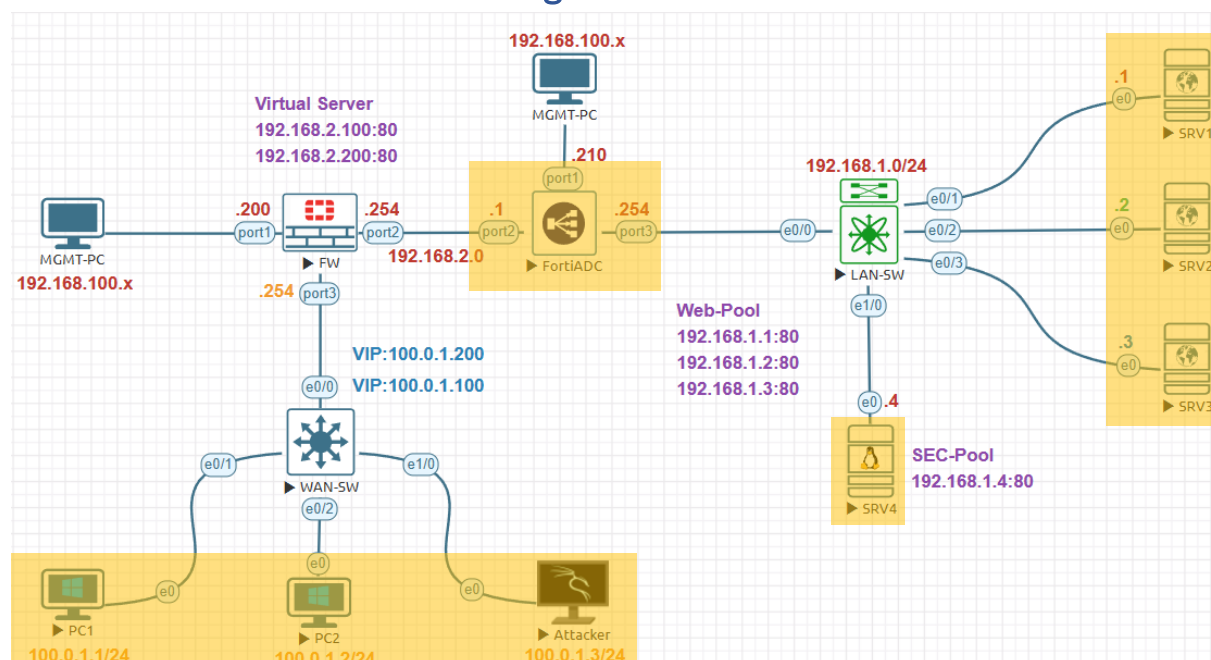


Least Connection Load Balancing Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Navigate to **Server Load Balance > Virtual Server** double click to open the setting.

Virtual Server

Name	Type	Address	Port
WEB-VS	Layer 4	192.168.2.100	80
SEC-VS	Layer 7	192.168.2.200	443

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

Go to **General** change the Method to **LB_METHOD_LEAST_CONNECTION** click **OK**.

Virtual Server

Basic General Security Monitoring

Configuration

Address: 192.168.2.100
Example: 192.0.2.1

Port: 80
Default: 80 Range: 0 or 1-65535. You can specify up to eight ports or port range: e.g., 80-90 100. Valid values are from 0 to 65535, with 0 for Layer-4 virtual serv

Connection Limit: 0
Default: 0 Range: 0-100000000 concurrent connections

Connection Rate Limit: 0
Default: 0 (disabled) Range: 0-86400 connections per second

Interface: port2

Resources

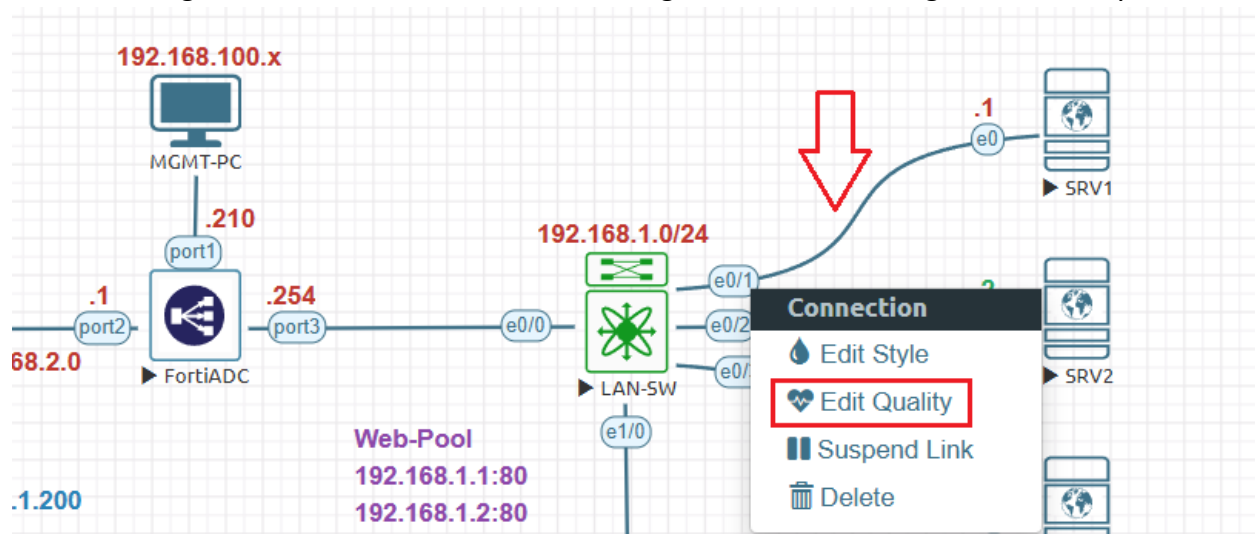
Profile: LB_PROF_TCP

Persistence: Click to select.

Method: LB_METHOD_LEAST_CONNECTION

Real Server Pool: WEB-Pool

Before testing the **Least Connection** load-balancing method, let's change SRV1 Quality

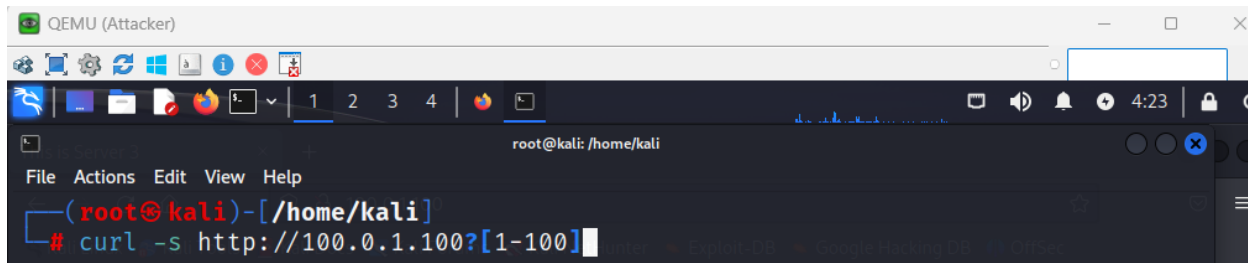


Change **Delay**, **Jitter**, **Loss** and **Rate** of the interface connected to LAN-SW e0/1.

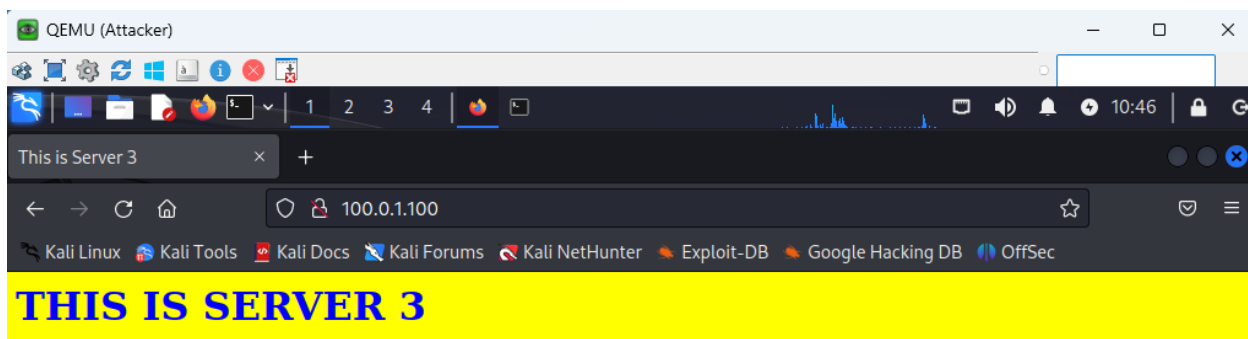
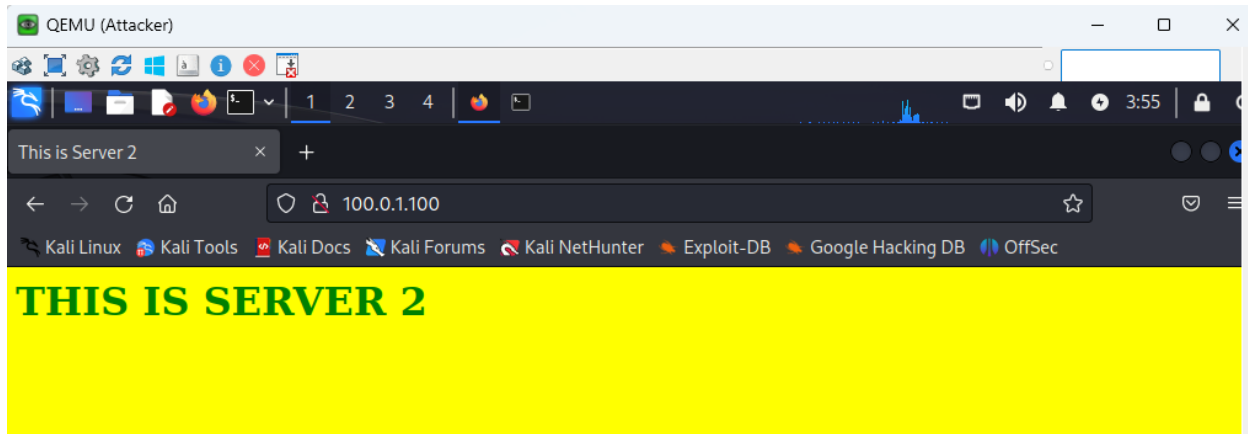
Link Quality: LAN-SW - SRV1					
Interface	Delay (ms)	Jitter (ms)	Loss (%)	Rate(kbps)	Interface
e0/1	90	50	40	10	e0

Test the virtual server by generating a significant amount of traffic against the virtual server. Open Attacker (Kali Linux) 100.0.1.3 machine open terminal and type below commands to generate traffic against the virtual server.

`curl -s http://100.0.1.100?[1-100]`



It hit Server 2 and Server 3 only because Server 1 response is very slow



Navigate to **Log & Report > Traffic Log > SLB Layer 4&2**

Source	Received Bytes	Destination	Sent Bytes	Service	Virtual Server	Duration (s)	Trans Destination	Real Server
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1245	tcp	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	1131	192.168.2.100	1244	tcp	WEB-VS	3	192.168.1.3	SRV3

Navigate to **FortiView>Virtual Server**

Total **2**
Healthy **2**
Unhealthy **0**
Unavailable **0**

Enable Analytics
Add Filter

☐	Name	Traffic Group	Health	Address	Port	RS	Pool	Type	Throughput (bits/sec)
☐	WEB-VS	default		192.168.2.100	80	3	WEB-Pool	L4	
☐	SEC-VS	default		192.168.2.200	443	1	SEC-Pool	L7	

Showing 1 to 2 of 2 entries
Show entries

Navigate to **FortiView>Virtual Server** click on **WEB-Pool** in Pool column.

Back to Virtual Server
Virtual Server: WEB-VS
Pool: WEB-Pool

Members
Health

Search

ID	Name	Status	Address	Port	Weight	Throughput (bits/sec)	Concurrent
1	SRV1	enable	192.168.1.1	80	1		
2	SRV2	enable	192.168.1.2	80	1		
3	SRV3	enable	192.168.1.3	80	1		

Showing 1 to 3 of 3 entries
Show entries